

COURSE TITLE : PRODUCT TESTING LAB
COURSE CODE : 6099
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/6
CREDITS : 3

TIME SCHEDULE

Module	Topics	Period
1	To arrive specific Documentation Of specification tests of various polymer products	25
2	Determination of mechanical properties of polymer products Determination of properties other than mechanical tests of polymer products	30
3	To analyse test data and correlated with standards & industrial needs	20
TOTAL		75

COURSE OUTCOME

Sl.	Sub	Student will be able to develop skill
1	1	Documentation Of specification tests of various polymer products
	2	Determination of mechanical properties of various polymer products and correlated with national standards
2	1	Analysis to test result data and correlated with specification values based on standards of various polymer products

LIST OF EXPERIMENTS

MODULE I

To arrive & Documentation of specification tests on the following polymer products

- 1) Cycle tyre & Cycle tube
- 2) Automobile tyres
- 3) Examination gloves
- 4) Surgeon's gloves
- 5) Latex foam
- 6) Electric cables.
- 7) Plastic containers
- 8) Some automobile components made from rubber, plastics and composites
- 9) Any other Polymer products having specifications.

MODULE II

Physical testing of Polymer Products

- 1 Determination of Tensile properties of rubber vulcanisate.
- 2 Determination of Tension Set.
- 3 Determination of Compression Set
- 4 Determination of Tear strength.
- 5 Determination of Hardness
- 6 Determination of Abrasion resistance
- 7 Determination of Resilience
- 8 Determination of impact strength
- 9 Determination of Mooney viscosity and cure characteristics
- 10 Determination of cure time using rheometer
- 11 Accelerated ageing testing.
- 12 Determination of cut growth resistance by the use of Ross flexing /De matia machine.
- 13 Determination of Tensile properties of plastics and composites.
- 14 Determination of Heat distortion temperature
- 15 Determination of flexural strength.
- 16 Determination of compressive strength.
- 17 Determination of MFI

MODULE III

- 1) To analysis of test data and correlated with standards & industrial needs